COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31s2
DATE PERFORMED:	23/07/2026	**DATE SUBMITTED:**	23/07/2026
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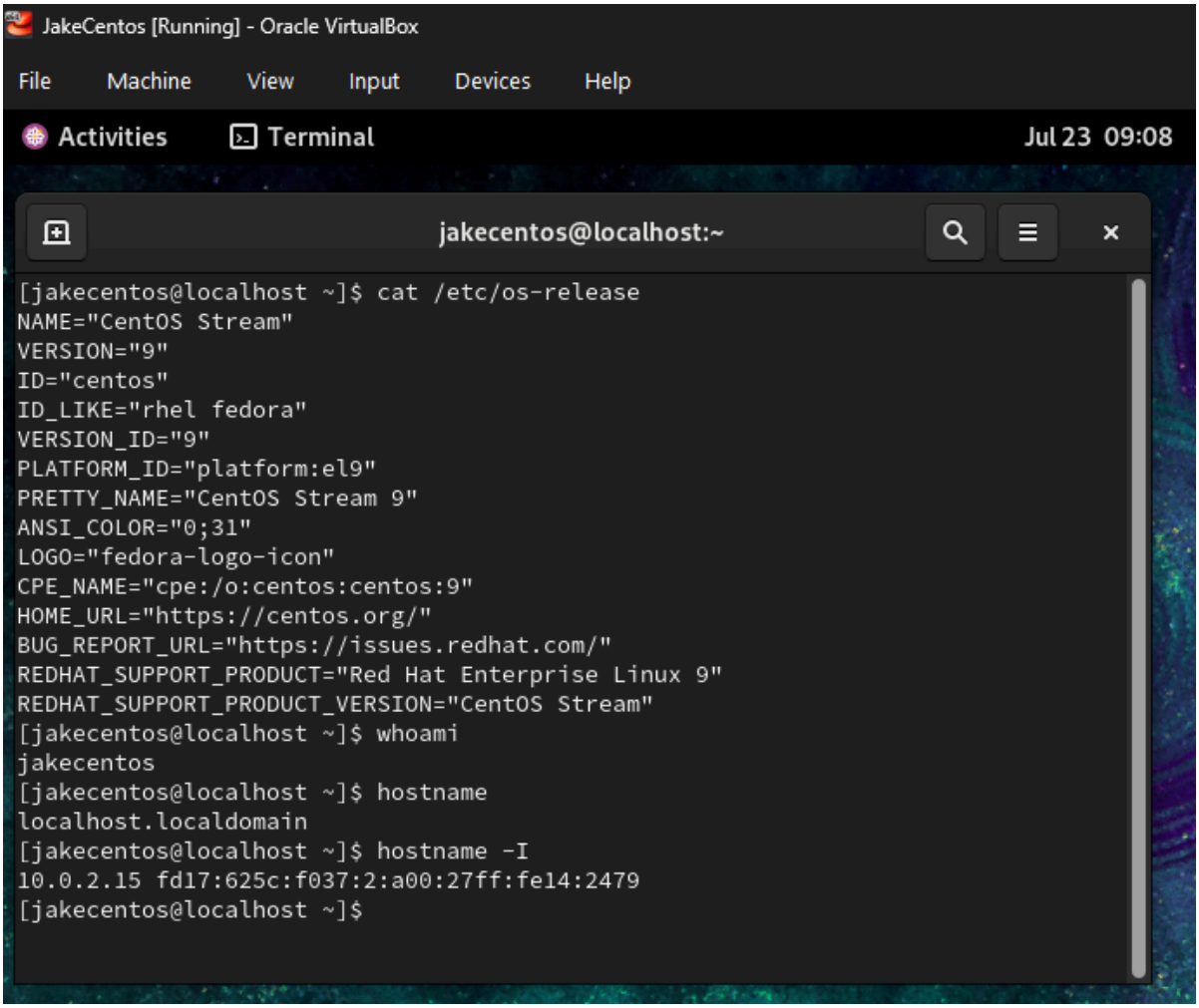
ACTIVITY NO. 3.1 Install SSH server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here: http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already



Task 2: Install the SSH server package openssh

1. Install the ssh server package openssh by using the dnf command: $ dnf install openssh-server
2. Start the sshd daemon and set to start after reboot: $ systemctl start sshd $ systemctl enable sshd
3. Confirm that the sshd daemon is up and running: $ systemctl status sshd
4. Open the SSH port 22 to allow incoming traffic: $ firewall-cmd --zone=public --permanent --add-service=ssh $ firewall-cmd --reload
5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes: $ systemctl reload sshd

```
[jakecentos@jakecentos ~]$ sudo firewall-cmd --zone=public --permanent --add-ser
vice=ssh
Warning: ALREADY_ENABLED: ssh
success
[jakecentos@jakecentos ~]$ sudo firewall-cmd --reload
success
[jakecentos@jakecentos ~]$
```

```
[jakecentos@jakecentos ~]$ sudo systemctl start sshd
[jakecentos@jakecentos ~]$ sudo systemctl enable sshd
[jakecentos@jakecentos ~]$ sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Thu 2026-07-23 09:31:09 PST; 6min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 952 (sshd)
      Tasks: 1 (limit: 24567)
     Memory: 2.1M (peak: 2.4M)
        CPU: 19ms
     CGroup: /system.slice/sshd.service
             └─952 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 23 09:31:09 jakecentos systemd[1]: Starting OpenSSH server daemon...
Jul 23 09:31:09 jakecentos sshd[952]: Server listening on 0.0.0.0 port 22.
Jul 23 09:31:09 jakecentos sshd[952]: Server listening on :: port 22.
Jul 23 09:31:09 jakecentos systemd[1]: Started OpenSSH server daemon.
[jakecentos@jakecentos ~]$
```

```
[jakecentos@jakecentos ~]$ ls -l /etc/ssh/sshd_config
-rw-------. 1 root root 3674 Jul 14 21:18 /etc/ssh/sshd_config
[jakecentos@jakecentos ~]$ sudo systemctl reload sshd
[jakecentos@jakecentos ~]$ hostname -I
10.0.2.15 fd17:625c:f037:2:a00:27ff:fe14:2479
[jakecentos@jakecentos ~]$
```

Task 3: Copy the Public Key to CentOS

1. Make sure that ssh is installed on the local machine.
2. Using the command ssh-copy-id, connect your local machine to CentOS.
3. On CentOS, verify that you have the authorized_keys.

```
jakeworkstation@WS:~$ ssh-copy-id -i  ~/.ssh/id_rsa.pub jakecentos@192.168.56.104
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/jakeworkstat
ion/.ssh/id_rsa.pub"
The authenticity of host '192.168.56.104 (192.168.56.104)' can't be established.
ED25519 key fingerprint is SHA256:stLsZl/Z2PvoRO9zZhbYqtaZIQQeRf0icMa3o8KbFyc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter o
ut any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompte
d now it is to install the new keys
jakecentos@192.168.56.104's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'jakecentos@192.168.56.104'"
and check to make sure that only the key(s) you wanted were added.

jakeworkstation@WS:~$
```

```
[jakecentos@jakecentos ~]$ ls -la ~/.ssh
ls: cannot access '/home/jakecentos/.ssh': No such file or directory
[jakecentos@jakecentos ~]$ ls -la ~/.ssh
total 8
drwx------.  2 jakecentos jakecentos   29 Jul 23 09:55 .
drwx------. 15 jakecentos jakecentos 4096 Jul 23 09:55 ..
-rw-------.  1 jakecentos jakecentos  753 Jul 23 09:55 authorized_keys
[jakecentos@jakecentos ~]$
```

```
[jakecentos@jakecentos ~]$ hostname -I
10.0.2.15 192.168.56.104 fd17:625c:f037:2:a00:27ff:fe14:2479
[jakecentos@jakecentos ~]$
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.
2. Show evidence that you are connected.

```
Last login: Thu Jul 23 09:44:13 2026
[jakecentos@jakecentos ~]$ whoami
jakecentos
[jakecentos@jakecentos ~]$ hostname
jakecentos
[jakecentos@jakecentos ~]$ cat /etc/os-release
NAME="CentOS Stream"
VERSION="9"
ID="centos"
ID_LIKE="rhel fedora"
VERSION_ID="9"
PLATFORM_ID="platform:el9"
PRETTY_NAME="CentOS Stream 9"
ANSI_COLOR="0;31"
LOGO="fedora-logo-icon"
CPE_NAME="cpe:/o:centos:centos:9"
HOME_URL="https://centos.org/"
BUG_REPORT_URL="https://issues.redhat.com/"
REDHAT_SUPPORT_PRODUCT="Red Hat Enterprise Linux 9"
REDHAT_SUPPORT_PRODUCT_VERSION="CentOS Stream"
[jakecentos@jakecentos ~]$ hostname -I
10.0.2.15 192.168.56.104 fd17:625c:f037:2:a00:27ff:fe14:2479
[jakecentos@jakecentos ~]$ S
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Reflections:
Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?
   - One thing the decision should include is determining the primary function of the computer, the kind of support provided its hardware and software reliability, the necessary software and the level of experience of the administrator.
2. What are the main differences between Debian and Red Hat Linux distributions?
   - The difference lies mainly in package management and system design. Debian packages are DEB and Debian systems are managed with APT. In comparison, the Red Hat family of systems use RPM packages and YUM/DNF for managing software.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini and Grammarly

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: For command assistance and grammar correction

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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